



Investigation of main bearing operating conditions in a three-Point mount wind turbine drivetrain

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Abstract

Wear-related failures of spherical roller bearings in the main bearing position of three-point mount wind turbines have been higher than expected and can contribute to higher than anticipated operation-and-maintenance costs. In this paper, the operational conditions of such a main bearing—including measured axial displacement and velocity subject to the estimated axial loads—are described for an instrumented, commercial wind turbine. The field measurements suggest a maximum axial speed between the bearing rings that is less than 2 millimeters per second. It is estimated that the axial speed between the ring and rollers is approximately 25% of this value. When compared to a speed of rolling from approximately 200–352 millimeters per second, the measured axial sliding is therefore significantly less than 1% of the speed of rolling. Previous numerical studies of lubricant film formation in rolling contacts have shown that the effect of axial sliding starts to be noticeable only when this ratio exceeds 10%; therefore, the axial velocity represents only a small disturbance to the nominal pure rolling case, and the influence on oil film building can be neglected. A simple analytic model of the main bearing motion was also developed and demonstrated similar displacement and velocity characteristics.

Untersuchung der Betriebsbedingungen des Hauptlagers in einem Dreipunkt-Windkraftanlagen Antriebsstrang

Zusammenfassung

Verschleißbedingte Ausfälle von Pendelrollenlagern im Hauptlager. Die Position von Dreipunkt-Windkraftanlagen war höher als erwartet und kann zu höheren als erwarteten Betriebs- und Wartungskosten beitragen. In diesem Artikel werden die Betriebsbedingungen eines solchen Hauptlagers - einschließlich der gemessenen axialen Verschiebung und Geschwindigkeit, die den geschätzten axialen Belastungen unterliegen - für eine instrumentierte kommerzielle Windkraftanlage beschrieben. Die Feldmessungen schlagen Sie eine maximale Axialgeschwindigkeit zwischen den Lagerringen vor, die weniger als 2 Millimeter pro Sekunde beträgt. Es wird geschätzt, dass die Axialgeschwindigkeit zwischen dem Ring und den Rollen ungefähr 25% dieses Wertes beträgt. Im Vergleich zu einer Walzgeschwindigkeit von ca. 200–352 Millimetern pro Sekunde beträgt das gemessene axiale Gleiten daher deutlich weniger als 1% der Walzgeschwindigkeit. Vorherige numerische Studien zur Schmierfilmbildung bei Rollkontakten haben gezeigt, dass der Effekt des axialen Gleitens erst dann spürbar wird, wenn dieses Verhältnis 10% überschreitet; Daher stellt die Axialgeschwindigkeit nur eine geringe Störung des nominalen reinen Walzgehäuses dar, und der Einfluss auf den Ölfilmaufbau kann vernachlässigt werden. Ein einfaches analytisches Modell der Hauptlagerbewegung wurde ebenfalls entwickelt und zeigte ähnliche Verschiebungs- und Geschwindigkeitseigenschaften.

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Fig. 1 Widespread wear damage to a main SRB. Photo courtesy of SKF

1 Introduction

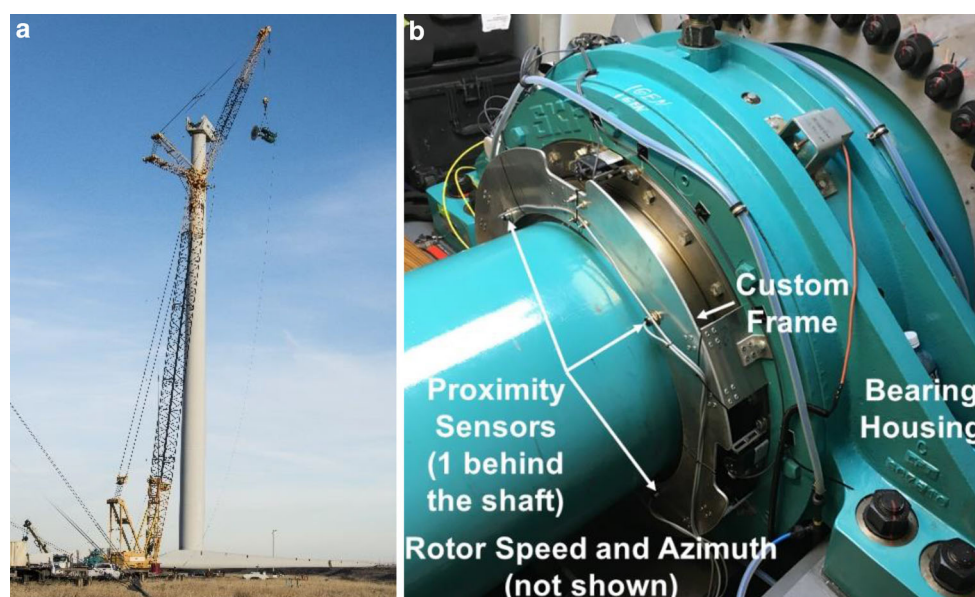
Wind power plant operational expenditures (OpEx) are an important but sometimes overlooked driver of overall leveled-cost-of-energy (LCOE) trends, remaining an appreciable contributor to LCOE with a capacity-weighted average of \$12 per megawatt-hour for recently commissioned land-based plants in the United States [1]. Recent data suggest that wind power plant OpEx can account for 25% to more than 35% of land-based and offshore wind LCOE [1, 2]. Turbine operation-and-maintenance (O&M) costs represent not only the single largest component of overall OpEx, and greater opportunities for future cost reduction likely exist within the turbine O&M category than within the many other components of non-turbine OpEx [1]. Drivetrain component reliability is an important contribution to turbine O&M and wind power plant LCOE in terms of both the expense of repairs and lost revenue during downtime [3].

The main bearing is a critical component for which replacement typically requires specialized, expensive, heavy-

lifting capabilities. A common drivetrain configuration in many land-based wind power plants is a “three-point” mount, in which a double-row spherical roller bearing (SRB) functions as the main bearing and primarily supports the rotor weight and forces, and the two gearbox torque arms support the torque and rotor moments. The three-point configuration is cost-effective, allows for misalignment, and is easy to maintain because it is modular; however, the failure rate for many of these SRBs has been higher than expected—often as high as 20–30% in as little as 6–10 years [4–6], though they were originally designed for a rating life exceeding 20 years [7]. Main bearing loads, including those induced by gravity and aerodynamic rotor thrust, are the result of interactions between the rotor and the complex wind field in which it is operating [8, 9]. Main bearing abrasion and wear, which is not accounted for in the design life estimation process, is one important potential root cause of failure [10], as shown in Fig. 1.

One popular hypothesis for the cause of this wear is that axial movement between the roller and raceway breaks down the lubricant film thickness in the contact [4]. In this paper, the axial displacement and velocity of a main SRB were measured and simulated during a variety of turbine operating conditions for an instrumented, commercial wind turbine. The maximum axial velocities were then compared to a predefined threshold to qualitatively determine the potential disturbance to lubricant film building. An analytic model of the main axial bearing motion was also developed and was used to examine the displacement and velocity characteristics.

Fig. 2 Drivetrain installation in GE 1.5-MW turbine (a) and main bearing instrumentation (b). Photos by Dennis Schroeder, NREL 49417, and Jerry Hur, NREL 49959



2 Experiments and instrumentation

In a cooperative research program, the National Renewable Energy Laboratory (NREL) and SKF have been investigating typical main bearing operating conditions since early 2018 [11]. An instrumented gearbox and main bearing were installed in a General Electric (GE) 1.5-megawatt (MW) SLE turbine, as shown in Fig. 2, and examined over a wide range of atmospheric and turbine operating conditions, including power production in normally occurring winds; parked and idling situations; and intentionally induced startup, shutdown, emergency stop, and grid events. The drivetrain measurements were time-synchronized with measurements from previously existing instrumentation on a meteorological tower in front of the turbine and on the turbine itself [12]. These measurements have also been supplemented by inspections of the bearing, including visual inspections of the rolling elements and raceways and collection of grease samples.

Fig. 2 also shows the housing of the BS2-8115/C2H main SRB and relevant instrumentation. Although similar to a standard 240/600 ECA/W33 double-row series SKF Explorer SRB with a bore diameter of 600 millimeter (mm), an outer diameter of 870 mm, a width of 272 mm, 28 rollers for each row, and a rated shaft speed of 18.3 revolutions per minute (rpm), its design was updated to optimize internal geometry specific to the turbine loads, to use a new cage design and material, to provide better sealing through a customized seal design, and to improve lubrication with an automated relubrication system [13]. Four inductive proximity sensors equally spaced around the circumference of a custom frame installed on the generator side of the main bearing housing measure the axial displacement of the main bearing locknut (and bearing inner ring) with respect to the main bearing housing (and bearing outer ring). When the main shaft and bearing inner ring move downwind, the distance between the locknut and the sensors and frame mounted on the bearing housing decreases. The proximity sensors were installed at the midpoint of their 10-mm measurement range relative to the locknut, and they have a repeat accuracy of ± 0.02 mm. This measurement method has the advantage that it is not affected by elastic deformation of the bearing housing or bedplate. Combined with measurements of the main shaft azimuth, the axial displacement can be resolved to account for out-of-flatness of the custom frame, locknut, and pivoting of the main shaft, which is essential to precisely determine the main shaft axial displacement [14, 15].

With the displacements recorded by these four inductive sensors, the axial displacement of the inner ring with respect to the bearing housing (representative of the bearing outer ring) can be determined through the following steps:

- Remove the spurious readings caused by a gap in the locknut.
- Estimate the readings in this region with a piecewise cubic Hermite interpolation.
- Convert to units of mm.
- Remove the influence of the misalignment and out-of-flatness of the retaining nut and locknut using measurements from slow rotations of the rotor.
- Apply a 2.5-hertz (Hz) low-pass filter to reduce noise.
- Average the four resulting signals, which remove any influence of pivoting of the shaft, to determine the main bearing axial displacement.
- Refer to the main bearing resting position by subtracting its displacement caused by gravity loads.

This process yields more accurate results than the simpler method used in previous work [11]. Once the main bearing axial displacement has been calculated, the main bearing axial velocity can then be estimated through time differentiation of it.

3 Analytic approach

The one-degree-of-freedom, analytic model of a main bearing in a three-point mount wind turbine drivetrain developed in this work describes the main bearing axial displacement subject to aerodynamic forces from the rotor. It is most suitable for normal power production cases and startups. The model can also be extended to simulate emergency stops by introducing a load correction factor. The model begins with the following assumptions:

- The system translates only in the axial direction; thus, the radial and pivoting motion of the main bearing along with the flexibility of the tower, bearing housing, main shaft, gearbox, and other elements are not considered.
- Damping is not considered, and time differentiation of the calculated axial displacement can be used to describe the main bearing axial velocity.
- The aerodynamic rotor thrust is the dominant force that causes the main bearing motion.
- The reaction loads at the main bearing and bushings in the gearbox torque arms are represented as spring stiffnesses acting in a parallel arrangement.
- The reaction loads from the generator coupling are not considered because the stiffness of this coupling is several orders of magnitude less than the stiffnesses of the main bearing and bushings in the gearbox torque arms.

A schematic of the drivetrain and relevant applied loads, gravity, and reaction loads are shown in Fig. 3. The degree of freedom of interest is the motion of the main bearing inner ring (and drivetrain) with respect to the fixed outer ring

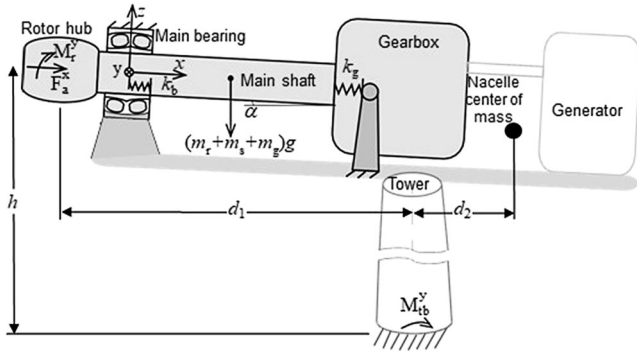


Fig. 3 Force diagram of the wind turbine system

in the axial (x) direction. The models are based on a force and moment balance of the drivetrain system, considering the system mass and weight, aerodynamic loads, and reaction loads provided by the interfaces between the drivetrain and bedplate.

The axial load applied to the main bearing, $F_a^x(t)$, and gravity loads resulting from the drivetrain tilt angle, α , are balanced by the spring stiffnesses of the main bearing, k_b , and each gearbox torque arm, k_g .

$$F_a^x(t) + (m_r + m_s + m_g) g \sin \alpha = (k_b + 2k_g)(x(t) + x_r) \quad (1)$$

This “resting” position under gravity, x_r , resulting from the masses of the rotor, m_r , main shaft, m_s , and gearbox, m_g , occurs when the rotor is stopped or slowly idling and the applied load can be assumed to be negligible.

$$x_r = \frac{(m_r + m_s + m_g) g \sin \alpha}{k_b + 2k_g} \quad (2)$$

The measurements described in this work are also most easily interpreted with respect to this resting position, the condition for which is much more easily determined than when the main bearing is at its geometric center; therefore, Eq. 1 can be simplified to determine the displacement of the main bearing inner ring (and drivetrain) with respect to the fixed outer ring, $x(t)$:

$$x(t) = \frac{F_a^x(t)}{k_b + 2k_g} \quad (3)$$

The main bearing axial stiffness, k_b , is a function of the contact state of each row of the bearing, which depends on the applied radial and axial loads. Given the applied radial load (i.e., mainly the rotor weight), the variation in stiffness with axial displacement from the geometric center of the bearing (i.e., $x = -x_r$) to the displacement in which one row is fully circumferentially loaded is shown in Fig. 4.

The stiffness is characterized by three separate contact conditions between the rollers and raceways, corresponding to the identified regions in the figure. In Region I, the bearing is near its geometric center, and both rows are partially loaded, leading to a relatively high axial stiffness. Under the slight axial load resulting from drivetrain tilt, as shown in Fig. 3, the main bearing displaces axially from its geometric center by the distance x_r , determined by Eq. 2, and rests approximately in the middle of Region I, as indicated in the figure. When the axial load increases enough to unload one bearing row, the bearing enters Region II. In this region, the bearing moves relatively freely within its operating axial clearance because the axial stiffness is considerably less. The operating axial clearance for the bearing is estimated to be 0.78 mm, which is approximately twice the displacement encompassed by regions I and II in the figure. That is, the clearance is typically defined as the displacement from one ring to ring, rather than from the center to the ring. When the axial load increases even more, such that one row is fully loaded circumferentially, the bearing is in Region III and the axial stiffness increases considerably.

Normal power production This model assumes that the aerodynamic rotor thrust is the only load applied to the main bearing. In a purely model-based scenario, simulation codes can be used to determine the aerodynamic rotor thrust. An alternative used in this work is to use measurements of the rotor pitch, M_r^y , and tower base moment, M_{tb}^y , to estimate the aerodynamic rotor thrust from a moment balance about the tower base:

$$F_a^x(t) = \frac{M_{tb}^y(t) - M_r^y(t) + (m_r d_1 - m_n d_2) g}{h \cos \alpha - d_1 \sin \alpha} \quad (4)$$

where d_1 is the overhung distance between the rotor hub and tower top centers, d_2 is the distance from the nacelle center of mass to the tower top center (positive downwind), h is the hub height (or distance between the tower bending moment location and the rotor hub), and m_n is the mass of the nacelle (including the gearbox and main shaft). The rotor pitch moment can be estimated from the blade root loads or those measured on the main shaft and expressed at the hub center. The tower base and rotor pitch moments in Eq. 4 will always contain some dynamic terms. In normal power production, these contributions are relatively small compared to the aerodynamic rotor thrust.

Emergency stop In an emergency stop, the aerodynamic rotor thrust quickly diminishes as the blades pitch and the rotor slows. The tower then begins to sway, so the inertia force quickly becomes the primary axial load applied to the main bearing. The axial load computed from Eq. 4 results in a significantly overestimated main bearing motion in an

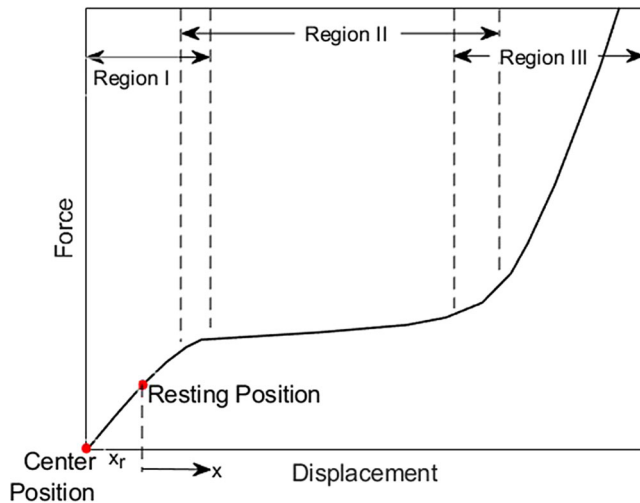


Fig. 4 Main bearing force-displacement curve

emergency stop because it represents the total system inertia. A load correction factor based on a mass ratio can be applied to the aerodynamic rotor thrust:

$$F_a^x(t) = \left(\frac{m_r + m_s + m_g}{m_r + m_n} \right) \cdot \frac{M_{tb}^y(t) - M_r^y(t) + (m_r d_1 - m_n d_2) g}{h \cos \alpha - d_1 \sin \alpha} \quad (5)$$

This force approximates the inertia from the rotor alone. The relevant modeling parameters for the turbine and main bearing are listed in Table 1; [16].

4 Results and discussion

In this section, the measured axial displacement and velocity of the main bearing is compared to the calculated results from the analytic model. A wide range of normal power

Table 1 Wind turbine analytic model parameters

Quantity	Symbol	Value
Distance from rotor hub center to tower top (m)	d_1	3.65
Distance from nacelle center of mass to tower top (m)	d_2	0.35
Hub height (m)	h	80
Gearbox bushing axial stiffness (kN/mm)	K_g	3
Gearbox mass (kg)	M_g	14,000
Nacelle mass (kg)	M_n	55,292
Rotor mass (kg)	M_r	31,700
Main shaft mass (kg)	M_s	5000
Drivetrain tilt angle (deg)	α	5

m meter, kN kilonewton, kg kilogram, deg degrees

production conditions are examined, along with a startup and an emergency stop. Finally, the potential impacts of this motion on main bearing health are discussed.

4.1 Normal power production

The cases examined in this section were gathered when the turbine was operating in normal power production conditions between March 2018 and November 2019. Nearly 400 data acquisitions limited to between 6 s (s) and 15 s, or approximately 2–5 shaft revolutions, in length were examined. A statistical summary of the measured electrical power and rotor speed along with the estimated aerodynamic rotor thrust is shown in Fig. 5. Data were acquired during almost the full power curve of the turbine, from very low wind speeds with light loads at cut in at a wind speed of 3.5 meters per second (m/s) to nearly the cut out at 25 m/s [16]. Above the rated wind speed, the rotor speed and power can vary by as much as $\pm 20\%$ from their rated values. The aerodynamic rotor thrust reaches a maximum of approximately 225 kN at wind speeds from 10–11 m/s and reduces as the wind speed increases beyond this point

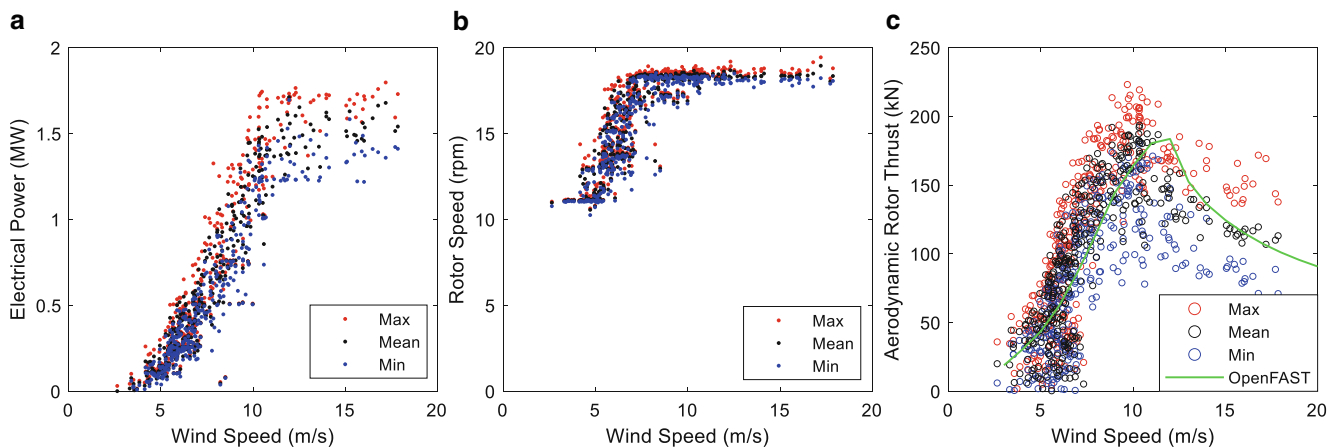


Fig. 5 Measured electrical power (a), measured rotor speed (b), and predicted aerodynamic rotor thrust (c) during normal power production

Fig. 6 Measured (a) and calculated (b) axial displacement during normal power production

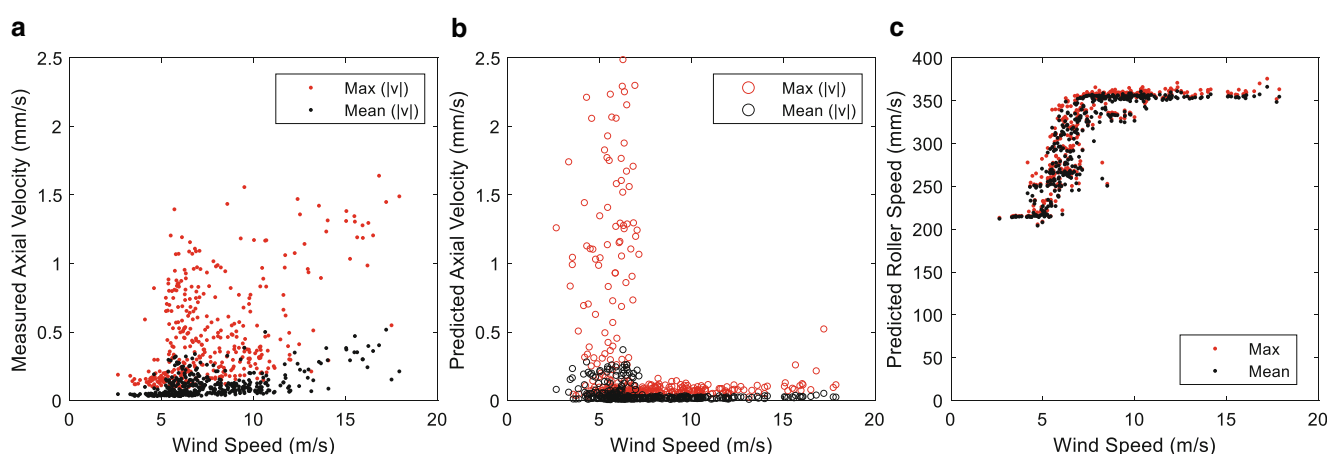
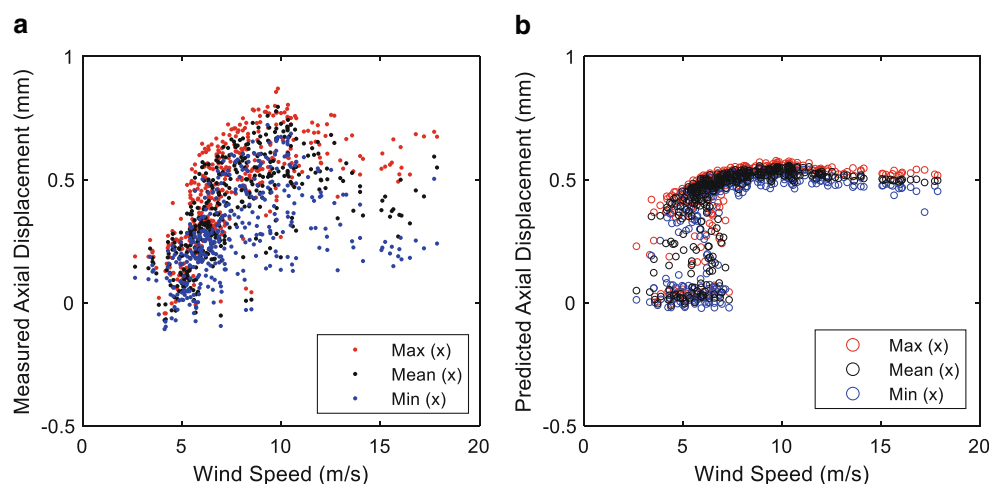
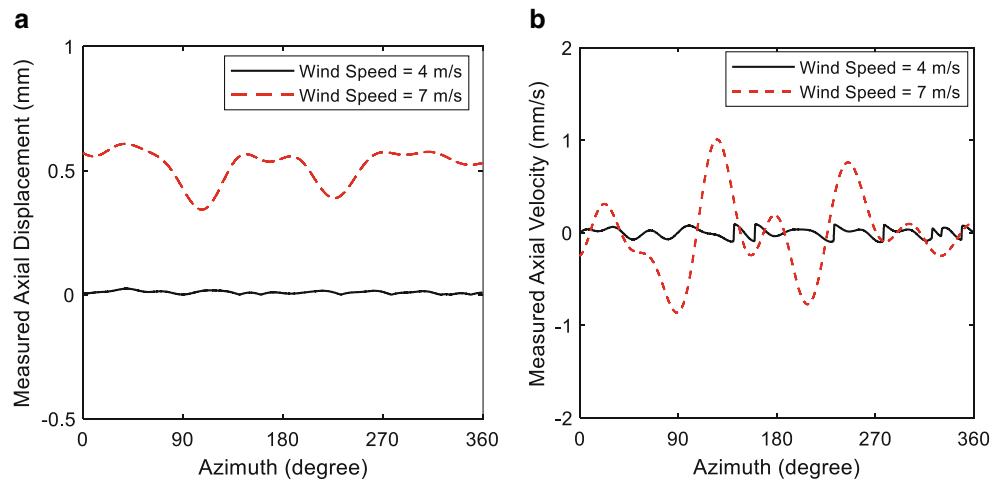
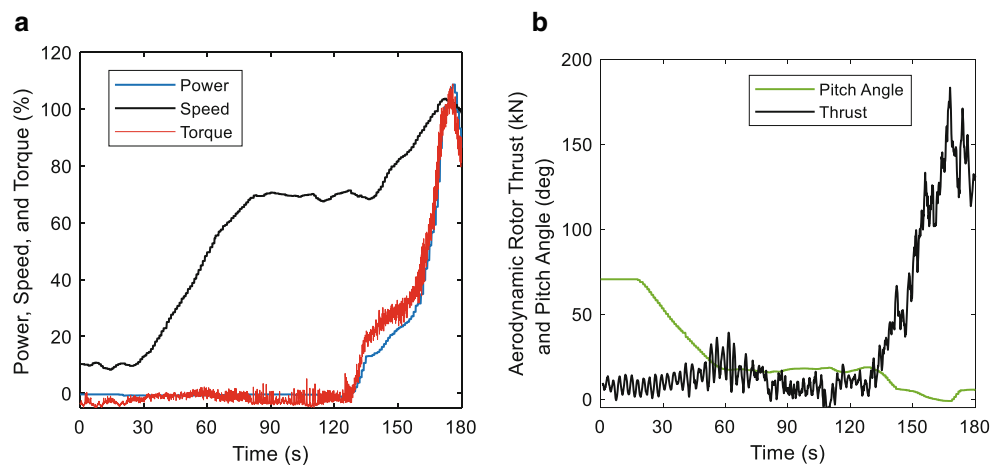
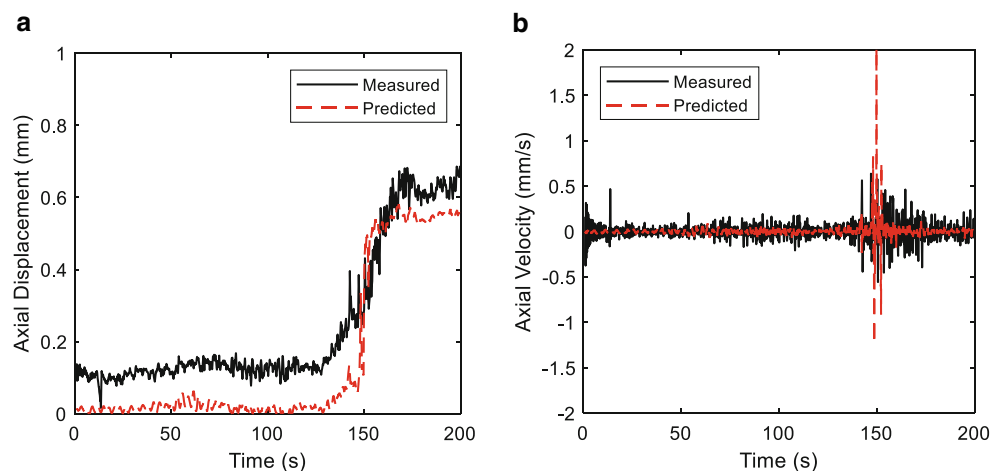


Fig. 7 Measured axial velocity (a), calculated axial velocity (b), and predicted roller speed (c) during normal power production

caused by blade pitching. The aerodynamic rotor thrust estimated from Eq. 4 is also compared to the prediction from the aero-servo-elastic simulation tool OpenFAST [17] using steady wind conditions. There is good agreement between the mean aerodynamic rotor thrust and the OpenFAST result.

The resulting main bearing axial displacement measured and calculated from Eq. 3 for these cases is shown in Fig. 6. The measured axial displacement generally follows the same shape as the aerodynamic rotor thrust curve, with the maximum axial displacement occurring at a wind speed of approximately 10 m/s. Overall, the main bearing moves downwind up to 0.85 mm from its resting position under gravity, x_r , illustrated in Fig. 4. This displacement includes movement through the axial clearance and into Region III. At wind speeds greater than 10 m/s, the aerodynamic rotor thrust and the axial displacement decreases; however, the model estimates fairly constant displacements. This suggests that the Region III stiffness used in the analytic model is higher than it is in reality.

A statistical summary of the absolute values of the measured and calculated main bearing axial velocity is shown in Fig. 7. The average velocity is less than 0.5 mm/s, but it can reach as much as 2 mm/s in some cases. Note that the turbulence intensity for this wind site varies tremendously—as high as 45% at low wind speeds and dropping to an average of approximately 15% at higher wind speeds [16]. Ambient temperatures at the site also can range from approximately -20 – 35 °C, which can change the temperatures of and the resulting temperature gradients between the bearing rings, main shaft, and housing. As a result, the bearing operating axial clearance is expected to vary to some extent during long periods of normal power production and throughout the year. At wind speeds less than 8 m/s, the main bearing is operating in either region I or II of the stiffness curve. In these regions, relatively small changes to the aerodynamic rotor thrust can result in appreciable displacements and velocities—in this case, up to nearly 2 mm/s. The analytic model estimations compare favorably to the experimental results, but—like the displacements—the velocities are lower than measured at higher wind speeds. Finally,

Fig. 8 Measured axial displacement (a) and velocity (b) during normal power production**Fig. 9** Turbine operating conditions during startup**Fig. 10** Axial displacement (a) and velocity (b) during startup

these relative velocities of the inner and outer rings can be compared to the roller rotational speed, assuming no slip. Generally, the relatively large radial load on the main bearing resulting from the rotor weight should be sufficient to ensure that this condition is met within the load zone. The speed of the roller surface, rotating about the center of the roller, ranges from 200–352 mm/s, which is generally two

to three orders of magnitude larger than the relative axial velocity of the main bearing inner and outer rings.

Fig. 8 shows two example time histories of main bearing axial displacement and velocity during one rotation of the rotor. The displacement is relatively constant and the velocity is very low for the 4-m/s wind speed case when the bearing is in Region I. Conversely, for the 7-m/s wind

Fig. 11 Turbine operating conditions during an emergency stop

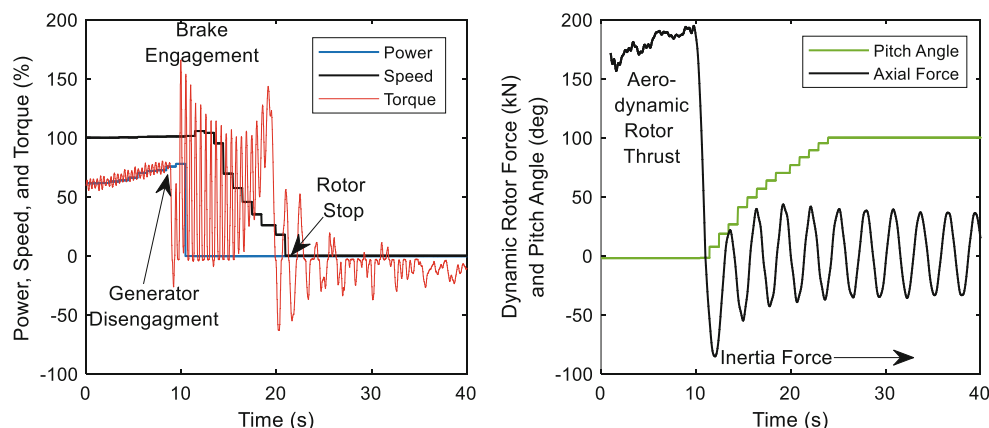
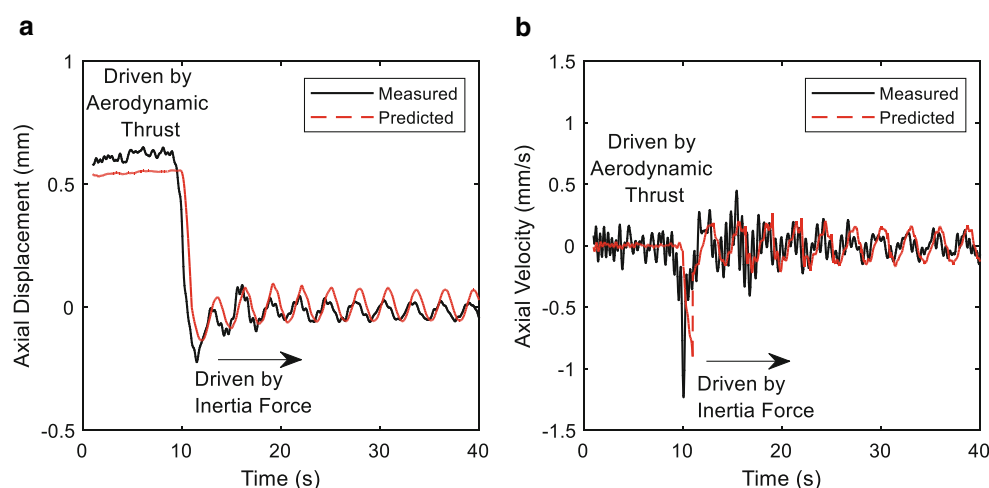


Fig. 12 Axial displacement (a) and velocity (b) during an emergency stop



speed case, the bearing is in Region II, and smaller changes in axial force can result in relatively large changes in displacement. For this case, the displacement varies to a much greater extent during the rotor rotation, resulting in velocities up to ± 1 mm/s.

4.2 Startup

The three phases during a startup event for this turbine are illustrated in Fig. 9. In the initial phase, the drivetrain speed increased from idle to approximately 70% (13 revolutions per minute) in the first 60s while under no appreciable torque. In the second phase, the turbine maintained this speed for approximately 60s until the generator engaged and the drivetrain began to transmit torque and generate electrical power. In the third phase, the blades reduced the pitch and increased the aerodynamic rotor thrust, torque, and drivetrain speed up to rated conditions. Additionally, this startup event occurred in March 2018, when the ambient temperature was lower than 0°C . The turbine idled for an extended time prior to the startup event, so the air temperature in the nacelle and the temperature of the bearing rings would also have been relatively and equally cold,

leading to slightly less axial clearance than in other situations.

The main bearing axial motion during this startup is shown in Fig. 10. Prior to engaging the generator, very little aerodynamic rotor thrust was produced, so the main bearing remained near its resting position in Region I until 120s. Some natural variation in the measurement of the resting position is evident from one case to another—in this case, the resting position is approximately 0.1 mm. After the generator engaged and the aerodynamic rotor thrust and torque increased, the main bearing relatively slowly and consistently moved downwind through Region II. After 160s and approximately 0.5mm of displacement, the main bearing was in Region III, and the displacement did not increase further. The main bearing axial velocity during the same startup is also shown in Fig. 10. The maximum axial velocity during the transition through Region II was less than 1 mm/s. The modeling results match the experimental data reasonably well because the dynamic motion of the system is relatively small compared to the aerodynamic rotor thrust.

4.3 Emergency stop

Emergency stops are one of the more extreme and dynamic events that wind turbines experience, an example of which is shown in Fig. 11. Initially, the turbine was operating at rated speed and 75% power. At 10 s, the emergency stop was intentionally induced, the generator was disconnected, and the reaction torque on the gearbox and main shaft was lost. The blades also began to pitch, and the mechanical brake on the gearbox output shaft engaged, dramatically increasing the gearbox torque to twice the operating torque just before the stop. The drivetrain then quickly decelerated from rated speed to stopped in approximately 10 s. During this period, there were many severe torque oscillations. After the rotor stops, the torque oscillations continued, but they slowly decreased in severity. The applied force in the axial direction predicted by the analytic approach and the corresponding pitch angle are also shown in Fig. 11. The aerodynamic rotor thrust dominates until the generator disconnection at 10 s. After the brake engagement, the load correction factor was applied to approximate the inertia force from the rotor.

Prior to the generator disconnection, the aerodynamic rotor thrust dominated the force applied to the bearing, just like the normal power production cases. As the rotor slowed and the blades pitched, the aerodynamic rotor thrust quickly diminished, and the inertia force dominated the force applied to the bearing. This inertia force is governed by the tower motion according to the first tower bending mode (0.35 Hz). The main bearing axial motion during the stop is shown in Fig. 12. Initially, the main bearing was displaced to approximately 0.5 mm in Region III. As soon as the brake was engaged, the rotor speed decelerated, the blades pitched, and the aerodynamic rotor thrust dropped quickly. As a result, the main bearing traveled upwind slightly past the resting position. The rotor and the tower swayed repeatedly, even after the rotor stopped, the effect of which can be seen in the main bearing displacement and velocity. The axial displacement and velocity calculated using the axial force shown in Fig. 11 agree well with the experimental results. Fig. 12 also shows the measured axial velocity during the emergency stop reaching a maximum of 1.3 mm/s, similar to the most dynamic normal power production cases. Note also that the bearing displacement is mainly described by a simple harmonic motion once the rotor fully stopped, after 21 s. This denotes that the bearing stayed in Region I and did not experience axial sliding during the tower oscillations.

During an emergency stop, the rotor comes to a complete stop while there is some axial motion. Both bearing loads and sliding distance become critical for these pseudo dry-sliding conditions, which can result in smearing. The wind industry has been using a well-accepted rule that compares the maximum radial load to the bearing's static load rating.

If this load ratio exceeds 7%, smearing can appear. Studies at SKF have shown that no bearing surface damage occurs until the bearing radial load exceeds 12% of its static load rating [18]. Derived from experimental data, the maximum of the load ratio during the emergency stop equals 4.9% (given the static load rating of 17,000 kN), less than the 7% threshold.

4.4 Potenzial implications of axial motion on main bearings

Based on the observations in the previous section, this section discusses the most important factors relating to main bearing reliability and the relevance of the observed axial motion.

Lubrication of main bearings Rolling element bearings and wind turbine main bearings are intended to operate primarily with sufficient lubrication. For wind turbine main bearings, this is typically achieved by filling the bearing housing with a suitable lubricating grease. Normally the grease is regularly replenished, either manually or automatically, to maintain a certain grease presence over time.

Classic film-building formulae are available since the work of Hamrock and Dowson published in the mid-seventies [19–21]. These formulae show that the resulting oil film depends on the speed of the rolling or entrainment, oil properties, and temperature as well as the curvatures and the materials of the two bodies in contact. The influence of load is quite small.

Bearing operating conditions in the NREL turbine Because the main bearing is loaded with combinations of radial and axial load, the rollers will be loaded accordingly, and there will be elastic contact deflection. At low and moderate roller loads, the contact area between rollers and raceways will be elliptical, with the small dimension of the ellipse in the direction of rolling. At high load, the contacts will approach a line contact.

The roller rotational speed described in Fig. 7 is derived from the ball-spin frequency used in vibration monitoring. The BSF is the rotational speed of a rolling element around its own axis of symmetry. For the SKF BS2-8115/C2H bearing, the BSF is 5.41 times the main shaft speed [22]. At the rated main shaft speed, the rollers rotate at 99.6 rpm. In the absence of tangential sliding, this is consistent with a linear surface speed of 352 mm/s, as shown in Fig. 7. This is the speed that is relevant for film building.

The impact of axial motion on bearing film formation With some background, we can approach the impact of axial motion on film formation in rolling contacts. This topic has been studied internally by SKF, but very little is found in

Fig. 13 Inspection of main bearing downwind row. Photos by Mark Dunn, SKF USA Inc., NREL 63050 and 63052



the literature. Film formation for rolling contacts is a matter of entrapment of oil transported into the contact by the rolling action. As noted, the speed of rolling is a major factor for oil film thickness in established formulae. It is perhaps intuitively clear that if an axial motion occurred that significantly dominated the rolling action, oil transport into the contact and film formation could be impaired.

SKF has performed numerical studies of film formation in rolling contacts and found that the effect of axial sliding starts to be noticeable when the ratio of the axial speed to the speed of rolling exceeds 10% [23]. At a ratio exceeding 50%, significant reduction is found. The modeling considers kinematics, elastohydrodynamic lubrication theory, and advanced rheology models of the lubricant. The impact on oil film thickness is not only caused by the kinematics but also to rheological phenomena, such as shear thinning of the lubricant.

Comparison the measured data for the NREL turbine From the experimental data of the present field study, we have measurements of the main shaft axial position relative to a housing-fixed setup of inductive position sensors. This should represent the inner ring-to-outer ring axial displacement and velocity; however, the speed of rolling should be compared to the roller-to-raceway axial velocity.

SKF has done several simulations with its proprietary multibody BEARING Simulation Tool (BEAST) and found that typically a sudden movement axially between the inner ring and the outer ring is distributed to approximately 50% relative roller motion in relation to each ring [24]. Further, it has been found that of the potential axial sliding between the roller and the raceway, approximately 50% of the motion is taken by the roller, assuming a slightly skewed direction of rolling. Altogether, this means that it is reasonable to expect approximately 25% of the ring-to-ring axial speed to transfer to actual axial sliding of the roller to the raceway.

Now, to understand how close we are to ideal film-building conditions, the relevant axial speed to compare is the speed of rolling. The field measurements suggest a maximum axial velocity between rings on the order of 1–3 mm/s.

To study the film formation between rollers and raceways in view of the advanced BEAST simulations, we should consider approximately 25% of that. In other words, we are left with less than 1-mm/s axial roller sliding to compare with the speed of rolling, which is typically on the order of 200–352 mm/s. The measured axial sliding is thus not even 0.5% of the speed of rolling. Moreover, the maximum ratio of the axial velocity to roller speed was found to be slightly greater than 0.5% during turbine idling, which is still much less than the specified 10% threshold.

Clearly, this is a small disturbance to the nominal pure rolling case, and the influence of axial sliding on oil film building can be neglected. If considered as an angle between the nominal direction of rolling and the actual direction of relative movement including axial sliding, 1 mm/s versus 352 mm/s is consistent, with an angle of ~ 3 milliradian. This leads us to conclude that the low level of axial velocity found in the measurements and simulations of this work should have quite a small influence, if any, on lubricant film formation.

4.5 Physical inspections

During its first 2 years of operation, the main bearing accumulated 2279 h of power production operation in the turbine. During that period, it was physically inspected twice, and the grease was sampled approximately every 6 months. Borescope images of the load zone taken during the second year of operation showed that the rollers (running surface and ends), cage, and inner and outer raceways were all in good condition, as shown in Fig. 13. The grease samples also noted no anomalies.

5 Conclusions

This paper described the operational conditions of an SRB in the main bearing position of a three-point mount wind turbine drivetrain. A commercial wind turbine was instrumented to measure the relative axial displacement of the inner and outer rings of the bearing in a range of opera-

tional states. A simple analytic model to estimate the axial displacement was developed using the estimated axial force on the bearing derived from aerodynamic models or blade and tower load measurements. Axial velocity was determined through time differentiation of the displacement. The axial velocities were then be compared to estimates of the speeds of the rollers to determine the influence, if any, on the lubrication state of the bearing.

The measurements suggest a maximum axial velocity between the bearing rings that is less than 2 mm/s for frequencies less than 2.5 Hz. Previous work has shown that the axial speed between the ring and the rollers is approximately 25% of that (i.e., 0.5-mm/s axial roller sliding), compared to a speed of rolling on the order of 352 mm/s. The measured axial sliding is therefore less than 1% of the speed of rolling. Because previous numerical studies of lubricant film formation in rolling contacts have shown that the effect of axial sliding starts to be noticeable only when this ratio exceeds 10%, this is a small disturbance to the nominal pure rolling case, and the influence on oil film building can be neglected.

Beyond examining the main bearing displacement and velocity measured from the field experiments in this work, future work will examine the operational temperatures, the load distribution characteristics, and the electrical current measured by other sensors installed on the main bearing. Additional bearing modeling will include examination of the bearing contact stresses and roller slip characteristics.

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